

WASP-43b

R-0475

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-43_231216 · created 2026-07-20T03:57:00+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2460295.0149 +/- 0.0014 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.167 +/- 0.02
Transit depth (Rp/Rs)^2	2.8 +/- 0.66 [%]
Orbital Inclination (inc)	83.44 +/- 1.53
Airmass coefficient 1 (a1)	1.155 +/- 0.014
Airmass coefficient 2 (a2)	-0.11 +/- 0.12
Scatter in the residuals of the lightcurve fit is	1.22 %
Best Comparison Star	#3 - [286, 269]
Optimal Method	PSF photometry
Transit Duration (day)	0.0546 +/- 0.0042

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.167 ± 0.02 vs 0.1594 (deviation 0.38σ)
Duration fit vs published	0.0546 d vs 0.0512 d (deviation 0.8σ)
Mid-time O–C	2.4 min
Depth SNR	8.3
Residual scatter	1.22 %

Light curve

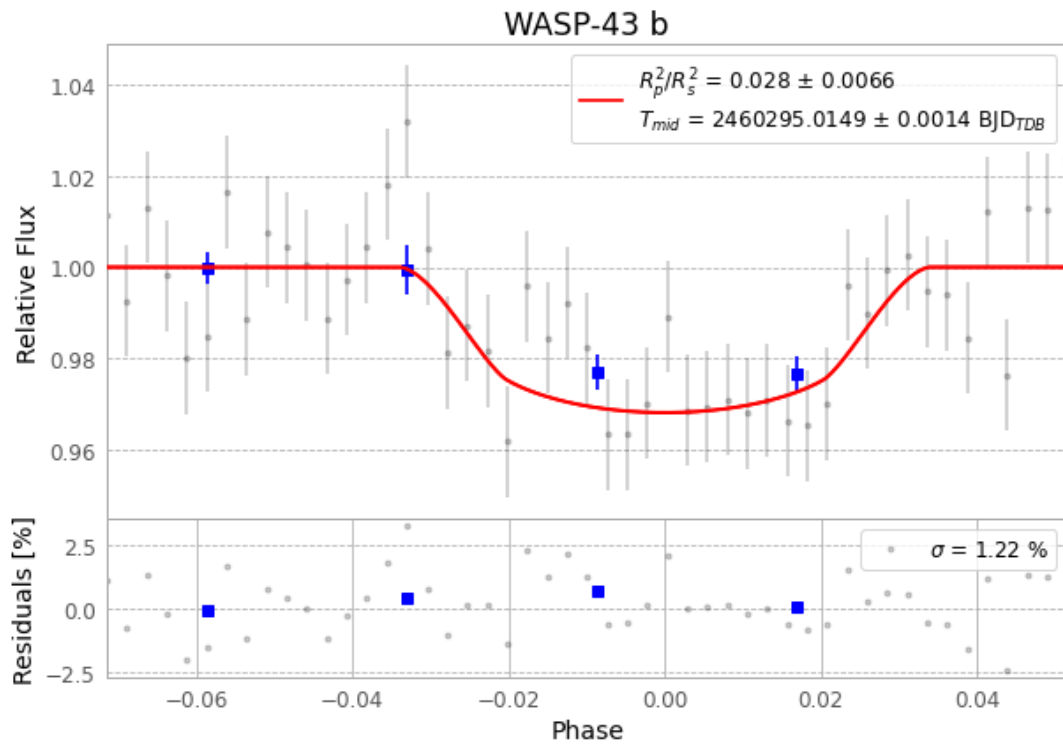


Figure 1 — FinalLightCurve_WASP-43 b_2023-12-16.png (SHA-256 8607672e063a8c9e...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [419, 223], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 65a10c56a4863940...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 39d9b80

Data provenance

49 FITS frames (32 MB), WASP-43231216105415.FITS → WASP-43231216131815.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-231216.log (a2f376f28d54...), FinalLightCurve_WASP-43 b_2023-12-16.png (8607672e063a...), FinalLightCurve_WASP-43 b_2023-12-16.csv (bdab89b48e7c...)

Compute resources

Wall time 150.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-20 03:57Z.